The virtual machine (VM) lifecycle refers to the series of stages a VM goes through from its initial setup to its eventual retirement. According to the sources, this process is essential for managing resources efficiently and maintaining system performance 1, 2.

Stages of the Virtual Machine Lifecycle

1. Creation and Provisioning

- **Creation:** This is the initial stage where a VM is instantiated from a template or a disk image 1. During this phase, administrators define key hardware parameters, including **CPU cores, memory (RAM), storage capacity, and networking configurations** 1, 3.
- **Provisioning:** Once created, the VM must be provisioned with resources. This involves attaching virtual disks, assigning **logical IP addresses**, configuring network settings (like VLANs), and installing necessary software components or operating systems 1, 4.

2. Deployment

Deployment involves placing the newly provisioned VM into its intended environment, such as **development, testing, staging, or production** 2. The strategy used depends on the specific use case and the workload the VM is expected to handle 2.

3. Operation and Monitoring (Running State)

- **Operation:** This stage covers the day-to-day management of the VM while it is in a running state. It includes applying security patches, updates, and troubleshooting issues as they arise 2.
- **Monitoring:** Continuous tracking of performance metrics—such as **CPU utilization, memory usage, disk I/O, and network traffic**—is required to identify bottlenecks or anomalies 5.

4. Scaling and Migration

- **Scaling:** If workload demands change, a VM can be scaled **vertically** (adding more CPU/RAM to the existing VM) or **horizontally** (adding more VM instances to distribute the load) 6.
- **Suspending:** In a **live migration**, the source host is temporarily **suspended** after its state is copied to the target host to allow the VM to move without losing its active state 7.
- **Migration:** This is the process of moving a VM's state from one physical host to another 2.
- **Live (Hot) Migration:** Performed while the power is **ON**. It requires shared storage to synchronize memory and state between hosts with minimal downtime 7.
- **Cold Migration:** Performed while the power is **OFF**. Configuration, log, and disk files are moved to the target host, resulting in longer downtime 7.

5. Decommissioning (Shutdown and Deletion)

- **Shutdown:** VMs may be shut down during off-peak hours to save costs or as part of a cold migration 7, 8.
- **Deletion (Decommissioning):** When a VM is no longer needed, it is retired. This final stage includes **archiving data, reclaiming physical resources** (CPU, RAM, storage), and completely removing the VM from the virtual environment 5.

Diagram Explanation

A lifecycle diagram typically follows a **circular or linear flow**:

1. **Start: Creation** (defining parameters) $\rightarrow$ **Provisioning** (allocating resources).
2. **Transition: Deployment** (moving to production).
3. **Active Loop: Operation & Monitoring** $\leftrightarrow$ **Scaling/Migration** (active running state).
4. **End: Decommissioning** (Shutdown $\rightarrow$ Data Archive $\rightarrow$ Resource Reclamation/Deletion).

10-Mark Exam-Ready Structured Notes

- **Introduction:** Define VM lifecycle as the management process of a VM from instantiation to retirement. Mention it is key to **resource optimization** and **high availability** 9, 10.
- **Phase 1: Setup:** Detail **Creation** (template-based) and **Provisioning** (software and network setup) 1.
- **Phase 2: Active Life:**
- **Deployment:** Mention different environments (Dev/Test/Prod) 2.
- **Running State:** Focus on **Operation and Monitoring** of metrics like CPU and Disk I/O 2, 5.
- **Dynamics:** Explain **Scaling** (Horizontal vs. Vertical) and **Migration** (Live vs. Cold) as methods to avoid failure and balance loads 6, 7.
- **Phase 3: Termination:** Detail **Decommissioning**, emphasizing the importance of **reclaiming resources** to prevent idle hardware 5, 11.
- **Practical Examples:** Mention **Docker commands** for lifecycle management, such as docker run (Creation/Deployment), docker stop (Shutdown), and docker rm (Deletion) 12, 13.
- **Conclusion:** Summarize how a well-managed lifecycle supports **cloud computing**, **disaster recovery**, and **cost efficiency** 10, 14.